

| Settings      | Visual Navigation |              |                      | Visual Tiling | Natural-Language Navigation |
|---------------|-------------------|--------------|----------------------|---------------|-----------------------------|
|               | Route Planning    |              | Next Step Prediction |               |                             |
|               | Completing Rate   | Succ Rate    |                      |               |                             |
| GPT-4 CoT     | 37.02             | 9.48         | 48.61                | 54.15         | 54.00                       |
| GPT-4 w/o Viz | 37.17             | 10.28        | 48.49                | 46.98         | 35.50                       |
| GPT-4V CoT    | 33.36             | 5.65         | 46.59                | 49.62         | /                           |
| GPT-4 VoT     | <u>40.77</u>      | <u>14.72</u> | <u>55.28</u>         | <u>63.94</u>  | <b>59.00</b>                |

Table 1: Performance of different GPT-4/4V settings in all tasks. Underline denotes statistical significance with  $p < 0.05$  when comparing GPT-4 VoT against all baselines using two-sample z-test, while  $p < 0.16$  is observed compared with GPT-4 CoT in natural language navigation task.

## 5 Analysis

As explained in section 3, one of the core aspects of VoT lies in enabling LLMs with visual state tracking. During the experiments, it was observed that GPT-4 CoT occasionally exhibited this reasoning pattern across several tasks with exception of route planning. Besides, incorrect visualizations of VoT are commonly observed in model outputs. In this section, our analysis of VoT primarily focuses on three questions: (1) Do visual state tracking behaviors differ among prompting methods? (2) How visualizations enhance final answers? (3) Can VoT benefit less powerful language models?

### 5.1 Do visual state tracking behaviors differ among prompting methods?

For each model output, we extract the sequence of visualizations sampled prior to generating the final answer and discard any visualizations generated thereafter. Then we compare the sequence length  $l_v$  with the number of reasoning steps  $l_s$ . We calculate Complete Tracking  $\sum_i^n (l_v == l_s) / n$  when a visualization  $v_i$  corresponds to each state  $s_i$ . Similarly, we calculate the Partial Tracking metric as  $\sum_i^n (l_v > 0) / n$  when at least one visualization is present before the final answer is generated. Figure 5 shows the significant differences between these settings. In the GPT-4 CoT setting, it demonstrated noticeable tracking rate across almost all tasks except route planning. This observation implies that **LLMs inherently exhibit the capability of visual state tracking when spatiotemporal simulation is integral to reasoning**.

On the other hand, the visual state tracking behavior is **sensitive to prompts** to varying degrees. As showcased in Figure 8 in appendix, after removing "reasoning" from the prompt of VoT, the visualizations are sampled after GPT-4 generates the wrong answer. Consequently, explicitly prompting LLMs to visualize their reasoning traces with **VoT markedly improves the visual tracking rate**, thereby enhancing overall performance. The potential contribution of code pre-training to this emergent capability is further explored in Appendix C.

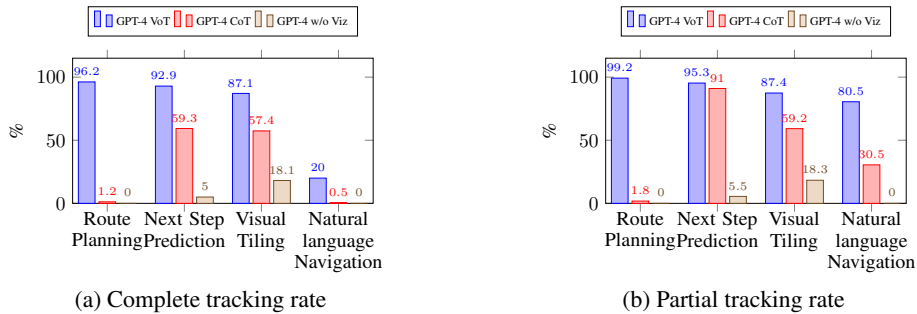


Figure 5: tracking rate of different settings across all tasks.

### 5.2 How visualizations enhance final answers?

Ideally, **VoT** is supposed to generate an accurate visualization  $v_i$  at each step, so that subsequent step  $z_{i+1}$  could be determined correctly. This relies on the spatial visualization and spatial understanding capability of LLMs. To evaluate these capabilities of LLMs in these tasks, we extract the final visualization from each model output under the setting **GPT-4 VoT** in visual navigation and polyomino

tiling task. Specifically, for visual navigation task, we extract the visualized map where LLM completed all navigation instructions. For polyomino tiling, we extract the rectangle filled with corresponding polyomino piece. The spatial visualization capability is measured by two criteria: (1) **Compliance**, indicating whether the manipulation of *mental image* satisfies requirements such as avoiding overlap and navigating around obstacles. (2) **Accuracy**, indicating whether the *mental image* aligns with the corresponding state. The spatial understanding capability is measured by the proportion of correct answers when the corresponding visualization is generated accurately.

As could be seen from Table 2, LLMs demonstrate promising potential in performing multi-hop visualization while adhering to spatial constraints, with compliance rates of approximately 51-52%. However, the relatively low accuracy of state visualization (around 24%-26%) indicates a need for significant improvements in this area. Despite this limitation, **LLMs are able to make correct decisions in 65%-77% of the cases when accurate internal state visualizations are generated**, which enhances groundedness and contributes to notable performance gains. Several case studies are provided in Appendix E for interested readers.

| Task              | Spatial Visualization |              | Spatial Understanding |
|-------------------|-----------------------|--------------|-----------------------|
|                   | Compliance            | Accuracy     | Accuracy              |
| Visual Navigation | 51.14                 | <b>26.48</b> | 65.16                 |
| Visual Tiling     | <b>52.01</b>          | 24.25        | <b>77.20</b>          |

Table 2: Spatial visualization/understanding evaluation in visual navigation and visual tiling task.

On the other hand, **VoT prompting might underperform in those tasks where LLMs can leverage logical reasoning without visualizing internal states**. We conducted experiments in natural language navigation within a ring [YBL<sup>+</sup>23], where navigation instructions are either clockwise or counter-clockwise movements. By normalizing each instruction to a signed number, GPT-4 converts this task to mathematical calculation of adding and modulus operation. For example, instructions of 15 steps clockwise and 3 steps counter-clockwise are normalized to  $(15 - 3) \% 12$ . Results show that GPT-4 CoT outperforms GPT-4 VoT with 52.5% VS 49.5% among 200 test instances with ring size of 12.

### 5.3 Can VoT benefit less powerful language models?

To evaluate the efficacy of VoT on less powerful language models, we conducted experiments across various model families [BMR<sup>+</sup>20, OA<sup>+</sup>23, TLI<sup>+</sup>23] and model sizes, including **GPT-3.5 turbo**, **LLAMA3-8B-Instruct** and **LLAMA3-70B-Instruct**. We access GPT-3.5 via Azure OpenAI API with model version 1106-preview and apply greedy decoding to all models.

As shown in Table 3, within the same model family, performance improves across all tasks with increases in model size. **LLAMA3-70B VoT** significantly outperforms the baseline across all tasks except for visual tiling, where it aligns closely with results observed in GPT-4. This consistency suggests that VoT offers a scaling advantage when applied to more advanced models, markedly enhancing performance in larger models. In contrast, less capable models tend to rely on random guessing, especially in spatial reasoning tasks. For instance, in the route planning task, GPT-3.5 CoT often resorts to speculative responses, random guessing in nearly half of the instances, which leads to exhaustion of output tokens. While GPT-3.5 VoT effectively minimizes random guesses, such occurrences become increasingly rare with GPT-4 CoT as the model size expands. On the other hand, the reliance on random guessing introduces unpredictability in performance trends for less powerful models. It suggests their limitations in sustaining reliable reasoning processes across different difficulty levels. Details on performance trends are provided in Appendix D.

## 6 Related Works

**Spatial Reasoning over Text** Spatial reasoning and spatial language understanding [KPM20] in NLP domain mainly focus on semantic representation [CBGG97, Bat10, HK11], spatial information extraction [RMK18, KVOM11], learning and reasoning [KM15, SLYA17, KFP19]. Recent advancements have further explored spatial reasoning within the context of large language models (LLMs). To improve multi-hop spatial reasoning skills of language models, several works [MFNK21, MK22] proposed to pretrain language models with synthetic datasets. An increasing number of dataset were then developed to covers various type of spatial relations in 2D visual scenes [WBC<sup>+</sup>15, SZL22],

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|                | Completing Rate     | Succ Rate          |                      |               |                             |
| GPT-3.5 CoT    | 16.10               | <b>2.62</b>        | <b>17.42</b>         | 44.10         | 8.50                        |
| GPT-3.5 VoT    | <u><b>19.02</b></u> | 1.61               | 13.10                | <b>47.99</b>  | <b>9.00</b>                 |
| LLAMA3-8B CoT  | 4.65                | 0                  | <b>28.73</b>         | <b>47.24</b>  | <b>16.50</b>                |
| LLAMA3-8B VoT  | <b>4.97</b>         | <b>0.2</b>         | 26.75                | 46.73         | 15.50                       |
| LLAMA3-70B CoT | 19.90               | 2.62               | 49.01                | <b>56.41</b>  | 26.00                       |
| LLAMA3-70B VoT | <u><b>30.24</b></u> | <u><b>5.85</b></u> | <u><b>54.09</b></u>  | 56.03         | <b>32.50</b>                |

Table 3: Performance of VoT in GPT-3.5 and LLAMA3 models. Underline denotes statistical significance with  $p < 0.05$  compared to corresponding CoT baseline using two-sample z-test.

geometric patterns [Cho19] and 3D spatial information [AMKK21, HZC<sup>+</sup>23]. [FML<sup>+</sup>22] investigated spatial reasoning capabilities of transformer-based models in the UI grounding setting. On the other hand, some works adopted in-context learning, leveraging LLMs for general purpose reasoning to convert spatial information to logic forms [YIL23], or as a general pattern machine for sequence transformation [MXF<sup>+</sup>23]. Recently, several works focused on evaluating spatial reasoning of LLMs as cognitive capability on navigation [YBL<sup>+</sup>23] and planning tasks [MHV<sup>+</sup>24] among various spatial structures. While most existing works rely on linguistic semantics and verbal reasoning, and might not always necessitate spatial awareness, we propose to elicit mind’s eye of LLMs in spatial reasoning tasks with various formats from a cognitive perspective. The VoT prompting induces LLMs to create *mental images* for visualizing their internal states and inform subsequent reasoning step.

**World Models of LLMs** While there have been many theoretical debates about whether LLMs can effectively learn an internal world model from ungrounded form alone [BHT<sup>+</sup>20, MGSS21], [LeC22] advocated that world models should represent percepts and action plans at multiple levels of abstraction and multiple time scales, with the capability of planning, predicting, and reasoning. [LWG<sup>+</sup>22] proposed to ground LLM in the physical world by reasoning over the experimental results predicted by external simulation. [HGM<sup>+</sup>23] further leveraged LLMs as world models to predict the subsequent states by action simulation, given predefined states and actions per task. On the other hand, an increasing number of studies focus on investigating internal representations of LLMs. [PP22, AKH<sup>+</sup>21] showed that by utilizing in-context learning, LLMs’ learned representations can be mapped to grounded perceptual and conceptual structure in color and spatial domains. Moreover, [GT23] and [NLW23] discovered linear representations of space, time and game state in specifically trained LLMs, which are important for dynamic causal world models. Our work does not probe the internal representations of specialized LLMs, nor does it depend on external simulation engine or state definitions. We demonstrate LLMs’ zero-shot capability of representing their precepts at an abstract level, predicting and tracking the internal states over time to generate action plans in multi-hop spatial reasoning tasks, which possibly mirrors the causal world model within LLMs.

## 7 Conclusion

This study introduces Visualization-of-Thought Prompting (VoT), inspired by the human cognitive function of visualizing and manipulating mental images through the mind’s eye. We have demonstrated that VoT enables LLMs to exhibit the mechanism of "the mind’s eye", as evidenced by their performance in multi-hop spatial reasoning tasks and our comprehensive analysis of the reasoning traces. Remarkably, VoT enable LLMs to outperform state-of-the-art multimodal large language models (MLLMs) in the tested visual tasks. While VoT demonstrates impressive efficacy in LLMs, this emergent capability to create *mental images* to enhance spatial reasoning resembles the mind’s eye process, suggesting its promise in MLLMs.

Building on the success of experiments with GPT-4, we plan to investigate how VoT can further elicit "the mind’s eye" in MLLMs to enhance their spatial awareness. Additionally, our future efforts will focus on automatic data augmentation from real-world scenarios, aiming to identify effective methods for learning generalized internal representations of *mental images*. This will further improve the mind’s eye of LLMs, ultimately contributing to the advancement of their cognitive and reasoning abilities.